

Supporting Information

Null-Model Choice for Rule-Defined Pattern Counts in Clustered Longitudinal Sequences: A Case Study of Jewish Halachic Menstrual Anticipation Rules

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This document contains the exploratory between-woman pairwise dependence analysis referred to in Sections 3.4, 4.4 and 5.2 of the main article, and the full SHA-256 checksum of the released filtered data file `data/FilteredData.csv` (1,554 cycles, 118 women), for verifying file integrity: `508ee88efb18bcd29c7ed6f841827377bb8d72a8c7dd38fdeb40cf08691f7811`. It is secondary to the null-model comparison and does not bear on the primary question of whether within-woman ordering produces excess rule-defined events. Notation follows the main article; all computations are produced by `scripts/postprocess.py` and the accompanying notebook in the code repository.

S1. Exploratory Pairwise Dependence Between Pattern Types

S1.1 Question and expectations

The joint test of the main article treats the five counts as one vector. The pairwise analysis asks a different and secondary question — whether women who produce one pattern type tend to produce another — as a consistency check against the relationships derived from the pattern definitions (main article, Section 2.2). Those relationships lead to two expectations: a positive $\text{Haflaga} \times \text{Week-Dilug}$ association, which is partly an arithmetic consequence of containment (every Haflaga sequence at $H = 30$ contains a Week-Dilug event) rather than an empirical hypothesis, and a plausible but not entailed positive $\text{Haflaga} \times \text{Dilug-in-Dilug}$ association from polynomial proximity. No expectation is held for the other eight pairs, so the analysis is a consistency check, not a search.

S1.2 Methods and rationale

Unit of analysis: the woman. — Pattern events are defined within a woman’s sequence and cycles within a woman are not independent, so the woman is the only exchangeable unit for an association test. The analysis works with 118 per-woman summaries, not with 1,554 cycles.

Rates rather than counts. — A woman with 45 cycles has roughly nine times as many opportunities to produce *any* pattern as a woman with 5 cycles, so raw per-woman counts of two patterns are positively correlated through follow-up length alone. Dividing each count by the number of cycles removes the first-order part of this exposure effect. It does not remove it entirely: a rate from 5 cycles is far noisier than one from 45, the minimum run lengths (2, 3 or 4 cycles) mean short sequences have proportionally fewer eligible windows, and in these data the Haflaga rate itself correlates with follow-up (Spearman correlation with n_i : 0.25 for Haflaga, 0.13 for Week, 0.10 for Dilug-in-Dilug, 0.02 for Dilug, 0.00 for Week-Dilug). The permutation null below is therefore one of *marginal* independence between two rates, not of independence conditional on follow-up; Section S1.4 checks how much this matters.

Spearman rather than Pearson correlation. — Per-woman rates are bounded at zero, heavily zero-inflated (most women have no event of a given type) and right-skewed. Pearson’s coefficient would be dominated by the few women with several events; the rank correlation measures monotone association and is insensitive to the magnitude of those few values.

Permutation rather than asymptotic p-values. — With 118 women, many tied zeros and skewed rates, the usual t -approximation for the null distribution of r_s is unreliable. One rate vector is permuted across women 50,000 times (seed 17); this breaks any association between the two patterns while preserving both marginal distributions exactly, giving a two-sided Monte Carlo p-value (proportion of permutations with $|r_s^*| \geq |r_s|$, with the $(b+1)/(B+1)$ convention).

Holm correction over ten pairs. — Five patterns yield ten pairs, of which only two carry a structural expectation. Holm’s step-down procedure controls the family-wise error rate at $\alpha = .05$ over all ten and is valid under any dependence among the tests, which matters because the pairs share patterns.

Binary robustness check. — Because the rate metric depends on the exposure normalisation and on the few women with several events, the analysis is repeated with each woman coded 1/0 for ever having established each pattern. Each pair is then a 2×2 table tested with Fisher's exact test (conditional on the margins) and with a permutation χ^2 statistic (permuting one indicator across women), again Holm-corrected. The probability of ever observing an event also increases with follow-up, so this check addresses the choice of metric, not the follow-up dependence.

S1.3 Results

Table S1 and Figure S1 give the rate-based Spearman correlations. Haflaga $\times$ Dilug-in-Dilug is positive and survives Holm correction ($r_s = 0.27$, $p_{\text{perm}} = .004$, Holm $p = .038$): women with a higher rate of exact repetitions also tend to have a higher rate of second-order progressions, most of which ($|d| = 1$ in 16 of 27) are near-constant sequences. Haflaga $\times$ Week-Dilug is positive, as containment requires, but does not survive correction ($r_s = 0.23$, $p_{\text{perm}} = .011$, Holm $p = .10$); only 3 of the 26 Haflaga sequences are at $H = 30$, so the guaranteed part of the association is small. The remaining eight pairs are null (all Holm $p = 1$), including Dilug $\times$ Week-Dilug, whose weak negative sign ($r_s = -0.12$) is in the direction implied by their mutual exclusion but is not distinguishable from zero.

The binary (ever/never) version reproduces the ordering and the two smallest p-values (Haflaga $\times$ Dilug-in-Dilug: $\chi^2 = 9.99$, $p_{\text{perm}} = .004$, Holm $p = .042$, Fisher $p = .004$; Haflaga $\times$ Week-Dilug: $\chi^2 = 7.26$, $p_{\text{perm}} = .013$, Holm $p = .12$), so the findings do not depend on the rate normalisation or on the few women with multiple events.

S1.4 Robustness to follow-up length

Because both rates in a pair can depend on follow-up, the unrestricted permutation tests marginal rather than follow-up-conditional independence. Two checks, computed by the same script with the same seed, bound the consequence for the two structurally expected pairs. First, the partial Spearman correlation given n_i (correlation of the rank residuals after regressing each rank vector on the rank of n_i) is 0.254 for Haflaga $\times$ Dilug-in-Dilug (unadjusted 0.269) and 0.239 for Haflaga $\times$ Week-Dilug (unadjusted 0.231). Second, restricting the 50,000 permutations to

Table S1: Pairwise Spearman correlations between per-woman pattern rates ($N = 118$), sorted by p_{perm} (two-sided, 50,000 permutations); p_{Holm} across the 10 pairs. §: pair with an analytically expected positive association (main article, Section 2.2). Bold: $p_{\text{Holm}} < .05$.

Pair	r_s	p_{perm}	p_{Holm}
Haflaga $\times$ Dilug-in-Dilug§	+0.27	.004	.038
Haflaga $\times$ Week-Dilug§	+0.23	.011	.102
Dilug $\times$ Week-Dilug	−0.12	.206	1.00
Haflaga $\times$ Week	+0.10	.277	1.00
Week $\times$ Dilug-in-Dilug	+0.09	.326	1.00
Dilug $\times$ Week	+0.09	.337	1.00
Week $\times$ Week-Dilug	+0.06	.476	1.00
Week-Dilug $\times$ Dilug-in-Dilug	+0.07	.480	1.00
Haflaga $\times$ Dilug	+0.04	.668	1.00
Dilug $\times$ Dilug-in-Dilug	−0.02	.819	1.00

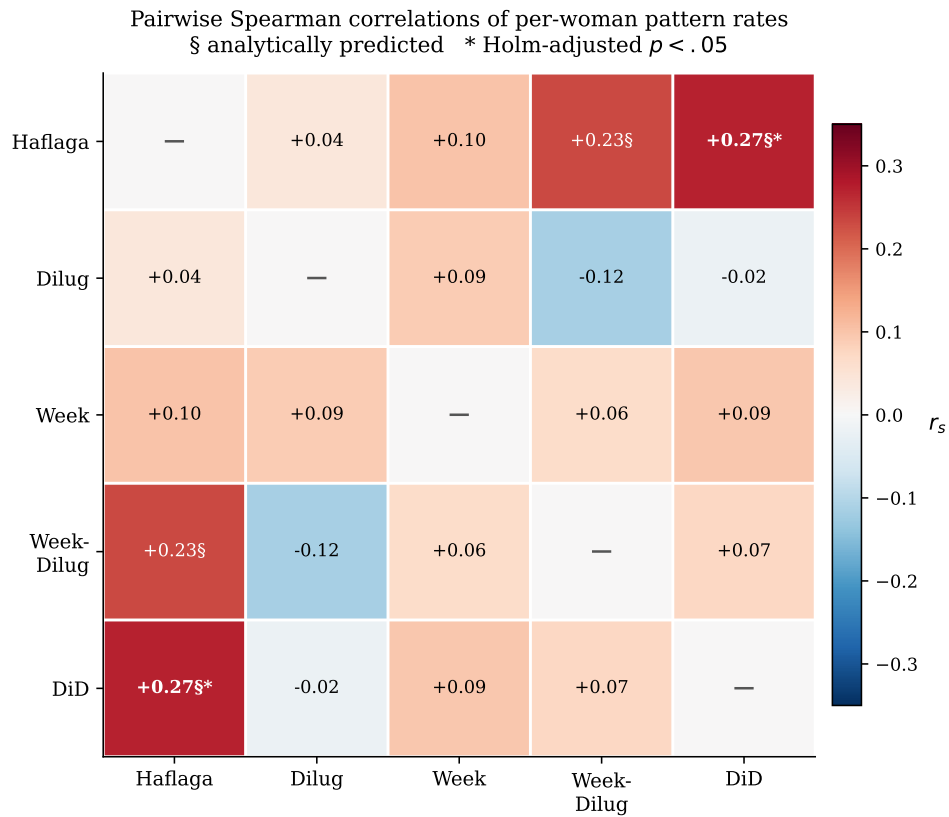


Figure S1: Spearman correlations between per-woman pattern rates. §: analytically expected positive pair; *: Holm-adjusted $p < .05$. All pairwise tests are exploratory.

exchange rates only among women in the same follow-up quintile (strata of 24, 8, 31, 27 and 28 women) preserves each rate's dependence on n_i while breaking the association between patterns; the resulting two-sided p-values are .0046 for Haflaga $\times$ Dilug-in-Dilug (unrestricted .0036) and .0097 for Haflaga $\times$ Week-Dilug (unrestricted .0113). The exploratory conclusions are therefore not an artefact of unequal follow-up. These checks are reported as robustness diagnostics for an exploratory analysis, not as an additional primary result.

S1.5 Interpretation

Three cautions apply. The associations are between women, not within sequences, and say nothing about temporal ordering. They involve rare patterns (18 Week-Dilug and 27 Dilug-in-Dilug events) and rest on few women. And the surviving association is compatible with the proximity argument but is not evidence for it: attributing co-occurrence to a mechanism would require an explicit generative model for the sequences, which the article does not fit.